Playing with Emotions: A Systematic Review Examining Emotions and Emotion Regulation in Esports Performance

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The massive growth of esports has vitalized the need to study human performance in competitive video gaming. The pressure of competitive play elicits a range of emotional experiences, which can affect players during and beyond a gaming session. In this work, we review the state of the literature concerning the role emotions play in esports performance as well as highlight coping strategies players use to regulate emotions during competitive play. We review the findings of N=32 peer-reviewed articles pertaining to emotions and esports, finding that the emotional experiences elicited by competitive play affect esports performance. In response, players attempt to regulate their emotions to maintain performance; however, efforts to do so vary, as they currently lack effective coping strategies. Lastly, we review the potential of technical interventions in esports training for improving emotion regulation among players. Our findings support knowledge development in esports, and present avenues towards promoting the emotional wellbeing of competitive gamers.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**; • **Applied computing → Computer games**.

Additional Key Words and Phrases: esports, performance, emotions, emotion regulation, affective computing

ACM Reference Format:
Nicole A. Beres, Madison Klarkowski, and Regan L. Mandryk. 2023. Playing with Emotions: A Systematic Review Examining Emotions and Emotion Regulation in Esports Performance. *Proc. ACM Hum.-Comput. Interact.* 7, CHI PLAY, Article 395 (November 2023), 30 pages. https://doi.org/10.1145/3611041

1 INTRODUCTION

The term 'esports' (electronic sports) refers to professional video game play performed at a competitive level, whereby expert players compete in organized sporting contexts (e.g., leagues, sponsored teams, or bracketed tournaments) in the presence of spectators [21]. Esports is a rapidly growing industry: revenues from esports were estimated at $24.9B USD in 2020 [2], and esports spectatorship is expected to grow to 646 million viewers by 2023 [63]. Within esports—and competitive multiplayer video games in general—players are under immense pressure to maintain performance, potentially eliciting a complex range of emotional experiences that invariably influence their performance within the game, their mood following the match, and their wellbeing in general. In such fast-paced environments where precise reaction times and optimal decision-making are crucial for determining the outcome of the game, a shift in emotion may significantly affect performance; developing effective coping strategies that esports athletes can use to manage their emotions before, during,

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2573-0142/2023/11-ART395 $15.00
https://doi.org/10.1145/3611041

and after gameplay is crucial for them to maintain optimal performance. Identifying emotional states that either facilitate or thwart performance could also inform the development of training interventions designed to foster emotion regulation strategies and maintain optimal performance. Additionally, professional esports players devote copious time and energy into improving performance, and if we can identify how emotions affect players, we may be better able to support the alleviation of symptoms of emotional distress—as well as to promote and maintain player wellbeing and performance through effective emotion regulation strategies [42].

In traditional physical sports, research suggests that performance outcomes are influenced by emotional states [38, 78]. In particular, studies have found that pleasant emotions (e.g., happiness) are associated with higher levels of performance, while unpleasant emotions (e.g., anger and anxiety) are associated with lower levels of performance [15, 59, 77]. However, we cannot assume—owing to the substantially different modality of play—that emotions experienced in the context of esports influence performance in the same ways as they do within traditional sports. Some research suggests that the competitive and cooperative nature of esports requires many of the same skills as traditional sports [14, 45], including mental skills like decision-making and coping with mistakes [27], teammate and performance-related stressors [55, 56], and emotion regulation skills [46, 48]. On the other hand, there is also evidence that the digital nature of esports introduces stressors not present in traditional sports, related to game design [36], technical difficulties [57], and the ubiquity of game-based toxicity [75]. Furthermore, contrary to research in traditional sports, research in affective computing has suggested that slightly elevated negative affect can be beneficial at work in myriad ways [69] that relate to performance in esports.

Despite the meteoric rise in esports, and the focus on emotions and emotion regulation in traditional sports, there is little research on the role that emotions play in esports performance, how players regulate their emotions, and what technological interventions are available to assist in this process. Supporting the emotional wellbeing of competitive game players, as well as facilitating expertise development, holds promise in providing better performance outcomes for all and in promoting the sustainability of esports. Additionally, assessing the potential of affective computing technologies in esports training interventions may contribute to the development of knowledge in the field of affective computing—and allow researchers to better understand how human affect and physiology influence esports performance. Finally, without an understanding of emotions in esports, it is difficult for esports organizations to support the implementation of training regimens designed to regulate emotions. Given these potential outcomes, it is important to review and assess the current state of the literature concerning the role emotions play in the performance and wellbeing of esports players to inform future research and help guide best practices.

While there exist several survey papers that review esports literature more broadly, and subdomains (such as stress and health) more specifically, none have as of yet reviewed emotion and emotion regulation literature in esports. To address this gap in the literature, this systematic review identifies the state of the literature concerning the role emotions play in esports performance as well as strategies for regulating emotions and maintaining performance. The first section provides an overview of how emotions influence performance in esports contexts—in particular, we identify what gameplay situations elicit emotions in esports and how performance is affected by such emotions. The following section reviews how players regulate their emotions during competitive play both on a team and on an individual level, as well as identifies emotion regulation strategies that may be implemented into esports training interventions. Additionally, we explore player perceptions of emotional control to gain insight into how mindset plays a role in esports performance and the effectiveness of emotion regulation strategies. Lastly, this review identifies both existing and potential technical interventions that may be used during esports training regimens to assist players in regulating their emotions. We generated three primary Research Questions:

- *RQ1.* What gameplay situations elicit emotions in esports?
- *RQ2.* How do emotions affect esports performance?
  - *RQ2a.* What is the effect of positive emotions on performance?
  - *RQ2b.* What is the effect of negative emotions on performance?
- *RQ3.* How do players regulate their emotions?
  - *RQ3a.* Does emotional control play a role in esports performance?
  - *RQ3b.* What coping strategies do players use both on a team and on an individual level?
  - *RQ3c.* What technical interventions exist to assist players in regulating their emotions?

A systematic review examining these questions better situates future research efforts in supporting and facilitating performance, as well as in understanding the interplay of performance and emotions in esports. In particular, we expect that answering these questions will help to equip researchers with a holistic understanding of emotional experience during competitive play—thus informing study design and useful paths of future inquiry. We also expect that a synthesis of extant literature will be advantageous to esports organisational staff, who may be able to utilise these findings (in particular, those concerning interventions) to better support rostered players' performance and wellbeing. In terms of player perspective, the ability to control one's emotions is considered a core game competency [46]—as such it is imperative to provide players with the tools required to reframe their emotions into an asset productive to their gameplay.

2 BACKGROUND

2.1 Extant Systematic Reviews in Esports

Within the last five years, researchers have conducted systematic reviews to outline the current state of esports literature published over the course of the past decade [5, 19, 50, 52, 62]. These reviews have been situated within various contexts, such as the psychological interest of esports (e.g., player characteristics and motivations) [5], the influence of esports on stress from a health perspective [50], esports trends across different disciplines of research [62], the cognitive and game performance of esports [52], and definitions of esports [19]. However, the aforementioned reviews do not explicitly address the role emotions have on esports performance. While Palanichamy [50] explore esports from a health perspective (e.g., physical issues, psychological distress, and addiction), their research questions pertaining to emotion focus exclusively on how esports impacts psychological health. Rather, our review identifies the multi-directional relationships between how emotion impacts esports performance and vice versa.

Furthermore, current systematic reviews frequently either intentionally exclude conference proceedings from their search [5], or potentially overlook proceedings within search strategies that do not include databases known for publishing such work (e.g., ACM Digital Library, DiGRA) [50, 52, 62]. We argue that, in the context of esports scholarship, this is an oversight. In the domains of Human-Computer Interaction and Games User Research (which are prolific contributors to esports research), conference proceedings are regarded as parallel to—and often likewise distributed in—journal formats, are relevant to both academia and industry, and are subject to the same standards of scientific rigour (vetted by blind peer review) as journals. As esports constitutes a highly interdisciplinary domain, we contend that all disciplines' publishing standards should be considered fully in review. Moreover, Pedraza-Ramirez [52] restricted their search to quantitative studies, excluding potentially relevant qualitative works. Considering esports is still a new and evolving field of research, qualitative works are particularly relevant in exploratory contexts and can provide a better understanding of the nuances embedded within problem spaces. The inclusion of both qualitative work and conference proceedings is in line with a recent systematic review of esports definitions by Formosa et al. [19]

We situate our review separately from existing systematic reviews in that we identified the role emotions play on esports performance and how players regulate their emotions. Furthermore, we expand our search strategy to include conference proceedings and databases known for publishing such work. Additionally, we include qualitative works within our review as they are equally valuable to our understanding of esports and emotion as quantitative studies. Moreover, self-reports are required to gain an accurate account of subjective experiences as psychophysiological measures do not reliably predict emotional states (see section 3.3). Taken together, we provide a more holistic review of existing works pertaining to the multi-dimensional role emotions play on esports performance and how players regulate their emotions.

2.2 Emotions, Emotion Regulation, and Performance

2.2.1 A Primer on Emotions and Emotion Regulation. While the terms 'affect', 'emotion', and 'mood' are often used interchangeably within research on games or esports, they are distinct concepts within psychological literature. *Affect*—or affective states—are the psychophysiological constructs that represent the domain relationship between mental states and physiological response, and are comprised of the joint experience of valence and arousal; for example, 'pleasure' and 'displeasure', or 'tension' and 'relaxation' [7]. In contrast, *emotions* (e.g., anger, pride, envy) contain both affect and other co-occurring components such as outward expression or behaviour, attention, and cognitive appraisal [65]. As such, emotion is complex—as it is higher level—and includes cognitive reflection of the experienced affect, learned cultural and contextual influences, and is labeled with language [7]. Meanwhile, *moods* are considered to be longer-lasting, more diffuse (i.e., not necessarily associated with a particular stimulus), and as biasing cognition more than action [25, 51]. Emotion researchers also consider the philosophy of emotion, the sociology of emotion, and the psychology of emotion with efforts to synthesize these interdisciplinary fields being an active area of research [7]. For the purposes of our paper, we are less concerned with any one specific epistemological approach to emotion, and consider *emotion as encompassing of affective responses to stimuli, that are culturally and cognitively interpreted, that may be mild or intense, and may be fleeting or longer lasting*. This would include emotions relevant to esports and competitive gaming such as feeling anger over a loss, frustration after missing a shot, joy after winning a match, boredom during training exercises, fear after exposure to game-based toxicity, jealousy when a teammate moves to a different team, or shame following underperformance.

Emotion regulation refers to "shaping which emotions one has, when one has them, and how one experiences or expresses these emotions"[24]. In this way, emotion regulation can be viewed as any approach that one uses to regulate (i.e., manage, cope with, or modify) their emotions, such as breathing deeply when stressed, calling a friend when sad, going for a walk after an argument with a spouse, listening to music when listless, or putting on sweatpants and watching Netflix after a long stretch toward a paper deadline. Because this categorization of emotion regulation is so broad, Gross [25] describes emotion regulation in terms of its three core features. First, there is the activation of a goal to modify the emotion-generation process. Second, there is engagement of the processes that are responsible for altering the emotion trajectory (which can be implicit or explicit, effortful or effortless, conscious or unconscious, and controlled or automatic). Third, there is the impact of emotion dynamics, which refers to the latency, rise time, magnitude, duration, or offset of the emotional response. In our paper, we approach emotion regulation using Gross's model, noting that *emotion regulation comprises these three core features—activation of a regulatory goal, engagement of regulatory processes, and modulation of the emotion trajectory*—which could include approaches used in esports and competitive gaming, such as going for a walk to shake off a loss between matches, listening to music that pumps you up before a match, breathing deeply during a respawn animation to reduce frustration, or reaching out to a teammate for support when feeling sad.

2.2.2 *Emotions and Performance in Esports.*

In traditional sports psychology, there is a known connection between emotion and performance with decades of empirical research exploring stress and coping in athletes (see Polman, 2012 [54]). Given the deleterious effect high stakes competition can have on performance, emotion regulation plays a critical role in maintaining performance [38]. Studies have found that athletes experience a range of competitive, organizational, and personal stressors under pressure [48, 67]—and failure to effectively cope with such stressors has been shown to negatively impact performance (see Nicholls and Polman, 2007 [48]). Within traditional sports literature, researchers widely adopt Lazarus's [38] cognitive-motivational-relational theory of stress—a model that has also been discussed in relation to esports performance to help conceptualize the ways in which players experience and cope with stressors across a variety of gaming genres and levels of play [39, 55, 58].

A growing body of esports literature has emerged within the past decade, paralleling similarities between traditional sports and esports. It has been suggested that the competitive and cooperative nature of esports requires many of the same mental skills as traditional sports [14, 45]. A study by Himmelstein et al. [27] found that the mental skills (e.g., knowledge of the game, decision-making, adaptive coping, etc.) and obstacles (e.g., inadequate coping strategies with anxiety, past mistakes, harassment, team conflicts, etc.) experienced by League of Legends players are similar to those experienced by traditional athletes. Comparatively, esports players experience similar teammate and performance related stressors as traditional sports [55]—with players adopting similar coping strategies across both contexts, such as problem-focused coping (e.g., problem solving, goal setting), emotion-focused coping (e.g., relaxation, acceptance) and avoidance coping (e.g., avoid the stressor) in an attempt to regulate their emotions [56, 57, 72]. In both contexts, emotion regulation plays a critical role in alleviating stress and determining performance outcomes—the ability to adopt effective coping strategies and overcome barriers of optimal performance in competitive play is essential for maintaining performance [46, 48].

While there is evidence pointing to the shared coping experiences among esports players and traditional athletes, esports differs from traditional sports in critical ways that extend beyond physicality. The digital nature of esports introduces unique sets of computer-mediated stressors—such as game design (e.g., changes in meta) [36], technical difficulties (e.g., disconnecting from game, unresponsive equipment) [57], and competition formats (e.g., limited rest between matches) [43]—that are absent from traditional sport settings. Moreover, a prevalent concern in computer-mediated contexts is that of online toxicity. While negative inter-player behaviours also manifest in physical sports, unique or severe permutations of toxicity in online gaming may present due to the disinhibiting effect online environments have on the behaviours of players [75]. The non-physical medium likewise centralises cognitive challenge: esports players require expertise specific to gaming contexts such as knowledge of ability usage and cooldowns, map layouts and specific terrain strategies, movement and positioning, and team-based mechanical synergies [18, 60]—all of which place high levels of cognitive demand on players, potentially diminishing their ability to effectively cope under pressure. In addition, the fast-paced nature of esports competition formats (distinct from traditional sports, in that there is little opportunity for rest between matches) reduces opportunities for players to effectively cope and regulate their emotions [43]. Finally, outside of organized professional contexts, players in competitive solo queue environments perceive stressors as both a threat and a challenge—likely due to uncertainty associated with competing on unfamiliar teams [56].

As such, we cannot wholly translate our understanding of emotion and performance from traditional sport contexts into the domain of esports. It is important to explore the role of emotion on esports performance within the digital domain to more confidently identify ways to help facilitate and support expertise development in esports.

3 METHODS

In conducting a systematic review, our goal is to synthesise the existing literature to better understand the role emotions play in esports performance. In doing so, we hope to inform future avenues of inquiry by outlining and collating the current state of the literature. To report our systematic review, we follow the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) 2020 Statement and item checklist [49]; see Figure 1. Per the PRISMA 2020 Statement, we use the recommended relevant checklist headings to report our methods below.

3.1 PRISMA Checklist

3.1.1 Eligibility Criteria. As the primary aim of this work is to review the current state of the literature concerning the role emotions play in esports performance, as well as how players regulate their emotions, there were no restrictions placed on the year of publication or the venue type; however, the current review does not include work published past August 30, 2022 (the point at which we concluded our search). Articles that extend beyond the scope of this review were excluded, such as those pertaining wholly to physiological response in esports (see section 3.3). As such, articles included in the review date back to as early as October 2012 and as late as August 2022 (see Figure 2 for a breakdown). The following inclusion and exclusion criteria were applied to studies identified in the search:

- Articles must concern both esports (either professional gaming, or gaming in a competitive multiplayer context) AND emotion.
- Studies pertaining to speedrunning were excluded (while speedrunning is arguably an esport, the context of competition is notably different from multiplayer titles).
- Articles specifically studying child populations (defined as populations under the age of 18) were excluded, due to the critical role developmental trajectory plays in emotions and emotion regulation among adolescent and child groups [66].
- Articles must constitute a peer-reviewed journal article or conference proceeding.
- Articles that used ‘mood” and ‘emotion’ interchangeably (consistent with our understanding of the terms) were included.
- Studies not available in English were excluded.
- Grey literature (e.g., technical documents, non peer-reviewed articles, blog posts, and newspaper articles) were excluded.

3.1.2 Information Sources. The literature search comprised peer-reviewed articles sourced from databases commonly used for publishing papers in the field of games user research: ACM Digital Library, DiGRA, Scopus, IEEE Xplore, ScienceDirect, Sagepub, and PubMed. Notably, our search strategy included literature from both journal and conference venues. To ensure we captured all relevant papers, we also searched both Google Scholar and reference lists of key papers for articles that may have been missed by the described databases. The search was undertaken between November 8, 2021 and August 30, 2022; literature published after this date is not included in the analyses.

3.1.3 Search Strategy. In a series of guided discussions, the authorship team collaboratively developed the final search string (with operators amended for unique database requirements), key terms,

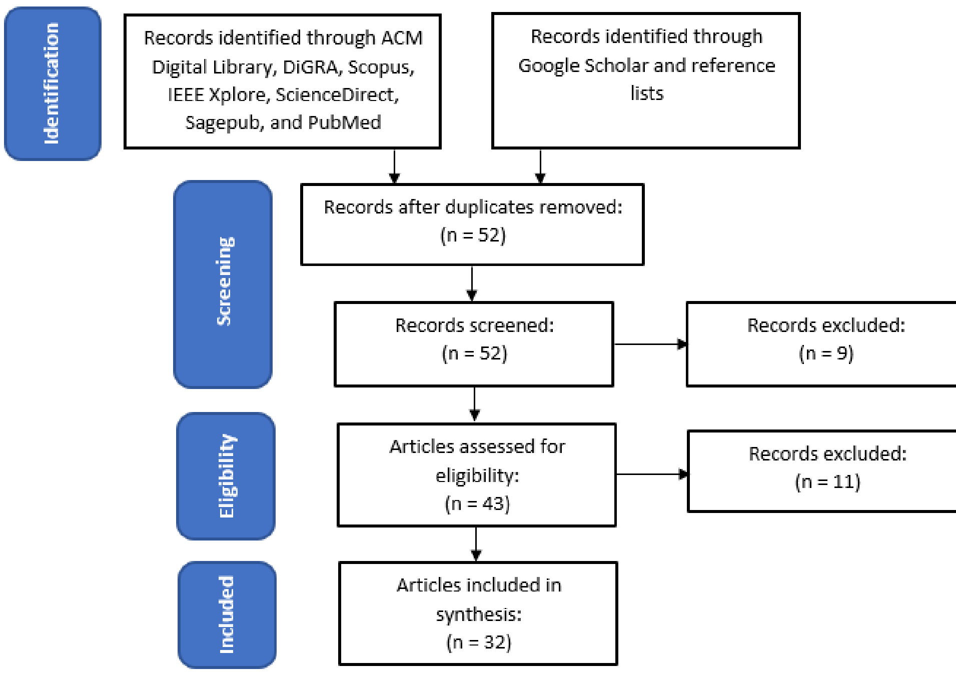

Fig. 1. Flow chart representing the systematic review process.

and proposed databases. A set of keywords and general terms relating to esports and emotion were used to extract all relevant articles from our set of databases. For example, the search string used for the ACM Digital Library database was: (("emotion" OR "mood") AND ("esport" OR "e-sport" OR "esports" OR DOTA* OR "CS:GO" OR "csgo" OR counterstrike OR "counterstrike" OR PUBG* OR playerunknown* OR Fortnite OR "lol" OR "Valorant" OR "R6 Siege" OR Overwatch or "Call of Duty" OR Hearthstone OR "Heroes of the Storm" OR "Halo 5")).

Game titles and their permutations were included to return searches that may have otherwise been overlooked in our primary search terms ("emotion", "mood", and "esports"). To ensure the inclusion of both previously and contemporarily popular esports titles, we identified the top 10 esports games—by tournament prize pool—of each year beginning 2017, and ending 2021. This resulted in a raw list of 18 games, including both a single mobile title ('Arena of Valor') and multiple individual entries for single franchises (e.g., Call of Duty: Warzone, Call of Duty: Infinite Warfare, Call of Duty: WWII, etc.). From there, we condensed all franchise titles and excluded the mobile title. The final list of games included were Defense of the Ancients, Counter-Strike, Player Unknown's Battlegrounds, Fortnite, League of Legends, Valorant, Rainbow Six Siege, Overwatch, Call of Duty, Hearthstone, Heroes of the Storm, and Halo. Included in the search string were common permutations of each game titles—e.g., 'LoL' for League of Legends, 'counterstrike', 'cs:go', and 'csgo' for Counter-Strike, and so on. It is important to note that while we did not explicitly search for other game titles beyond this list, it is still possible for papers about other titles to appear in our search of keyterms, such as "emotion", "mood", and "esports".

An initial pilot search of the ACM Digital Library was undertaken by all authors; all authors then independently reviewed the same set of collected literature, and excluded works that did not fulfill the inclusion criteria. Following consensus on excluded works in the initial review, the first author undertook a full search of all described databases.

3.1.4 Selection Process. Papers identified by the described search terms in the initial database search were screened by title and abstract. All authors participated in this process, reviewing a subset of the collected literature. Following this, the full corpus of collected literature was then subject to a full-text review to discern eligibility and relevance. Both the first and second author then reviewed the full-text of each article, meaning each paper received two independent reviews. This helped to ensure compliance with the eligibility criteria. Following both the initial screening process and within the eligibility review, a total of n=32 papers were found to be eligible for review (see Figure 1).

3.1.5 Data Collection Process. Data was extracted from the 32 papers with the use of a data extraction form. This form was developed, piloted, and populated by the first author. All data was extracted by the first author. Following this, both the first and second author independently reviewed the form. Conflicts in the assessment of core variables (e.g., a paper's key findings) were recorded individually by the second author during their review—these conflicts were then resolved through structured discussion (entailing conflict description, concurrent manuscript review, and consensus) following review. No action was taken to seek additional information from the paper authors—as such, all findings reported within this review pertain wholly to data reported in the published literature.

3.1.6 Data Items. Data items included the author(s), year of publication, research focus, methodology, and key findings of each paper. The research focus and key findings were later used to assist in the coding process, wherein each paper was assigned (non-exclusively) to one relevant category (as developed by the first author). The methodology was primarily used to assist in identification of application to the eligibility criteria (e.g., if the study complemented physiological evaluation with additional measurements), and to ground the findings in relevant contexts (e.g., the esports title used).

3.1.7 Synthesis Methods. No meta-analysis was conducted on statistical information, as we did not limit our review to purely quantitative literature. As such, synthesis of the results focuses on a discussion of the trends and relevant findings in the data extracted from the papers.

3.2 Coding

Following the literature search, articles were assigned (non-exclusively) to three categories—'emotion and esports performance', 'emotion regulation', and 'technical interventions'—based off the article's primary focus and research questions (see Tables 1, 2, and 3). These categories were developed by the first author, who likewise assigned each article non-exclusively to the relevant categories. These categories and assignments were then reviewed by the second author; consensus in paper categorisation was arrived at prior to reporting. Conflicts in categorisation were resolved via structured discussion, following a similar resolution process to that described in Section 3.1.5.

3.3 Physiological Literature

Our search strategy frequently returned literature that measured physiological response during esports play. While we did not explicitly search for terms that would return physiological inquiries (e.g., "heart rate", "EEG"), our keywords returned this work as a function of discussion: often, authors would speculate as to the possible emotional implications of the physiological data. Of particular focus in this space was the physiological arousal and stress that occur either throughout a play session, or during specific in-game events [10, 12, 35, 74]. As the relationship between physiological response and emotion is not one-to-one [13], and the authors of this literature consequently clarified in many articles that potential connections to emotion are speculative, we have only included in this review physiological work that attempts to corroborate physiological signals with subjective self-report measures of emotion. That is: physiological signals can rarely be directly interpreted as a discrete emotion (e.g., fear, sadness) *without* the aid of subjective report measures; for example, increased heart rate is only indicative of physiological arousal—and could imply any number of states, including fear, excitement, startle response, or even physical exertion. As such, we have excluded physiological literature that does not also seek to complement the identification of emotion through subjective report. However, understanding physiological signals in the space of esports and performance is an important and ongoing research inquiry, and a future and specific cohesive review of this particular topic would be beneficial for esports research and industry. This decision was guided by the second and third authors, who each possess expertise in physiological assessment in the context of games scholarship.

4 RESULTS

Collection of the full corpus of literature revealed 52 papers. Application of the exclusion criteria (see section 3.1.1) reduced this to 32 papers for inclusion in the final study (see Figure 2 for publications over years). The majority of the articles were related to emotion regulation in esports (n=15), while the remaining concerned emotions and esports performance (n=13) and emotion-centric technical interventions (n=4). Figure 1 illustrates a flow chart of our systematic review process. Tables 1, 2, and 3 show a summary of the literature concerning emotions and emotion regulation on esports performance.

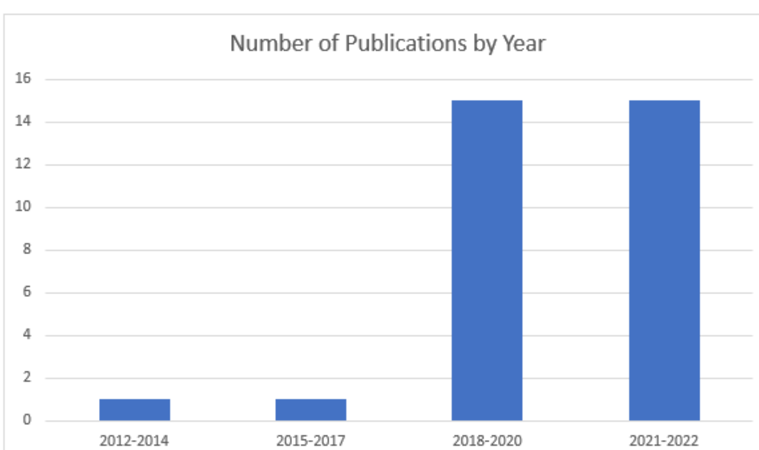

Fig. 2. Number of publications by year.

4.1 Category Overview: Emotion and Esports Performance

For ‘emotion and esports performance’ (see Table 1), 13 articles were selected from the literature search that focus primarily on the role emotions play on esports performance (RQs 1 and 2). In particular, these articles present findings concerning specific gameplay situations that elicit emotions in esports (RQ1), as well as how positive and negative emotions affect esports performance (RQ2)—revealing a nuanced understanding of the interplay between emotion and performance outcomes (see sections 4.2 and 4.3). Among these studies, two used quantitative measures (i.e., questionnaires), two used qualitative methods (i.e., survey responses), five employed mixed methods (i.e., questionnaires, and experiment), three were solely experimental, and one used machine learning (see Figure 3).

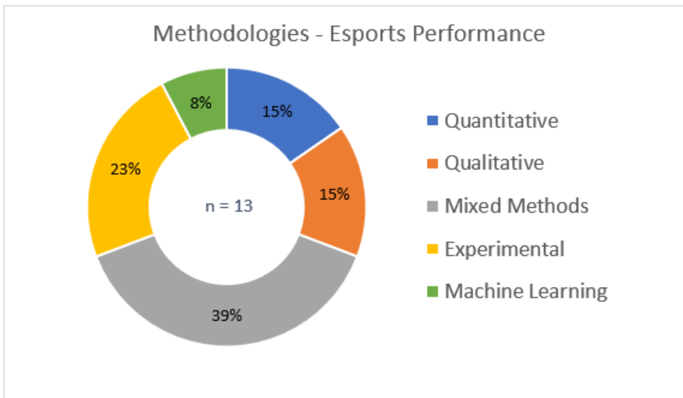

Fig. 3. Methodologies used in studies related to emotion and esports performance.

4.2 RQ1: What Gameplay Situations Elicit Emotions in Esports?

The following section breaks down the literature evaluating gameplay situations that elicit emotions into emotions experienced throughout stages of gameplay, emotive factors, and emotional responses to performance outcomes.

Stages of Gameplay. In the domain of esports, researchers have sought to understand what factors elicit emotions during competitive play. For instance, stress levels may vary among players throughout different stages of gameplay [26] and players may recall the peaks and the ends of a gameplay experience with greater emotional intensity [53]. In a similar vein, recent research among professional League of Legends players suggests that the pre- and post-game moods of players are associated with the outcome of the game [43]. Mateo-Orcajada et al. [43] found higher values of depression, anger, and post-game confusion following the loss of a game and higher values of vigor following a victory. Additionally, it was found that pre-game depression was significantly higher when the outcome of the last game was a loss; however, Mateo-Orcajada et al. [43] suggest this could be due to the competition format, which forces players to play consecutive games with limited rest between matches. Likewise, Shin et al. [71] explored both positive and negative emotions associated with *Counter-Strike* gameplay using electroencephalography (EEG) and self-report measures of emotion completed before, during, and after play. As indicated by EEG recordings and corroborated by self-report measures during gameplay, players experienced positive emotions associated with shooting players, and negative emotions associated with being shot. Subjective ratings of emotional response suggest players experience happiness during and after the game, whereas anger and arousal are only reported during the game.

Emotive Factors. Relevant to our research question, Kou and Gui [36] identified three emotive factors (i.e., situations that trigger players' emotional experiences) among League of Legends (LoL) players through an inductive thematic analysis of 431 posts on the LoL subreddit. The following factors are not mutually exclusive and can work together to influence a players' emotional experience:

- Achievement-related situations: moments in play involving achieving or failing to achieve objectives (e.g., defeating an opponent in combat, winning a game, or promoting in rank).
- Teammate-related situations: moments where the performance or behaviours of teammates trigger an emotional response (e.g., being flamed by teammates, or teammates failing to assist others in securing objectives).
- Game design-related situations: changes to the mechanics or content of the game itself (e.g., patches, updates, or specific champion nerfs and buffs).

In achievement-related situations, negative emotions such as anger and frustration were elicited when players felt their performance levels did not meet their own expectations. Meanwhile, players may experience positive emotions such as happiness and pride when they execute a skillful play. During teammate-related situations, negative emotions (e.g., frustration, anger, tilt) are generated when teammates underperform or engage in toxic behaviours (e.g., verbal harassment, trolling, griefing)—similarly, Poulus et al. [57] identified breakdowns in teammate communication as a major stressor among professional esports players. In contrast, positive emotions (e.g., feeling 'chill') occur when teammates are in harmony with each other and work together. In game-design related situations, players may experience sadness when their preferred champion is weakened (i.e., nerfed) or happiness when it is strengthened (i.e., buffed). A longitudinal analysis of stressors experienced by League of Legends players [58] echoes similar findings. In this study, Poulus et al. [58] collected diary entries of 6 elite LoL players over an entire competitive season (87 days). It was found that general performance, outcome, critical moment performance, and teammate mistakes accounted for the majority (55%) of the stressors reported. Furthermore, competition increased the severity of stressors—more stressors were reported in competitive diaries than training diaries, and the stressors were reported as more intense in the former.

Performance. Research by Mateo-Orcajada et al. [43] investigating emotional responses to performance outcomes found that heart rate and favourable performance actions (i.e., tower destruction, kills/deaths/assists (KDA), achievement of objectives) were positively correlated with vigor, and negatively correlated with negative emotions such as tension, depression, anger, fatigue, and confusion. However, when the performance actions of players benefited the opposing team (e.g., failure to secure an objective), heart rate and performance was negatively correlated with vigor, and positively correlated with the negative mood states. Likewise, research by Mavromoustakos-Blom et al. [44] found that certain facial expressions of emotion in Hearthstone (as corroborated by subjective response) are connected to physical interactions (e.g., eye gaze) with opponents—happiness is positively correlated with paying attention to the opponent, while opponent influence is associated with both happy, fearful, and angry facial expressions. For context, Mavromoustakos-Blom et al. [44] suggest that an in-game event that brings happiness to one player might evoke feelings of frustration and anger within their opponent.

In a similar vein, research by Behnke et al. [8] explored which situations within video game play elicit positive and negative emotions among gamers. Behnke et al. [8] analyzed 652 responses from Counter-Strike: Global Offensive (CS:GO) gamers using Linguistic Inquiry and Word Count (LIWC) to identify situations where they reported feeling amused, enthusiastic, angry, or sad. In amusing situations, gamers reported making mindless game mistakes or ridiculous shots during recreational play, while enthusiasm was reported during successful games in competitive settings. For situations that elicited anger, gamers often reported unfair situations such as hacking, cheating, trolling, or smurfing. Lastly, losing a game commonly evoked sadness among gamers. Interestingly, when referring to situations that elicit positive emotions (i.e., amusement and enthusiasm), gamers mentioned clutch performances such as victories and skillful plays. In contrast, negative emotions (i.e., anger and sadness) were described in relation to defeating scenarios such as their poor performance, weak teammates, or the unfair behaviours of others. This finding is in line with additional studies that point to how poor performance of the self or others elicits negative emotions among gamers such as anger and frustration [27, 37]. Likewise, negative emotions such as anger are expressed differently across game genres and player types—Balint et al. [3] found that while both Fortnite and League of Legends (LoL) players express their anger more passively (directed inwards) compared to non-gamers, Fortnite players use more passive expressions than LoL players likely due to less opportunities to communicate with others (e.g., no chat, limited voice options).

Summary. Taken together, the literature points to common themes concerning general gameplay factors, individual factors, and opponent factors that elicit emotions during competitive gameplay. In terms of general gameplay factors, negative emotions (e.g., depression, anger, frustration, tilt) are often associated with the loss of a match or objectives [8, 36, 43, 58], breakdowns in team communication and performance [36, 57, 58], as well as game design related changes that affect the viability of certain character picks or gameplay strategies [36]. Comparatively, positive emotions (e.g., happiness, pride, vigor) are associated with favourable gameplay factors such as winning a match [36, 43], securing objectives [43], character buffs [36], and cohesive team performance [36, 57]. With regards to individual performance, negative emotions such as tilt are often experienced when individual performance levels fail to meet a player's own expectations in competition [36]; however, this finding is contradicted by work that suggests one's own performance failures (e.g., missing shots) in recreational settings evokes feelings of amusement [8]. Lastly, opponents elicit negative emotions among players when opponents gain a competitive advantage (e.g., secure objectives, use cheats) [36, 44], deal damage to players (e.g., land shots or abilities) [36, 71], or engage in toxic behaviours [3, 36]. Overall, favourable gameplay situations generally elicit positive emotions while negative emotions are associated with the reverse; however, emotional response may be moderated by the context of play—with competitive settings evoking more negative reactions to unfavourable performance outcomes than recreational play [8].

The aforementioned literature has interesting applications to industry—knowledge of the types of gameplay situations that elicit emotion among players (both within and beyond their control) leads to a greater understanding of how it can be managed. For instance, a variety of negative emotions are elicited by factors outside of the player's control, such as poor teammate performance [8, 27, 37, 57, 58], changes in game design [36], tiresome competition formats [43], and opponent actions [36, 44]. Focusing on factors that are within a player's control, such as individual performance, may reduce the elicitation of negative emotions and lead to increased feelings of positive emotions (e.g., self-efficacy). As such, esports organisations and support personnel—such as coaches, managers, and other stakeholders—may benefit from considering these factors in training regimens and real-time coaching, and encourage players to direct their focus productively.

4.3 RQ2: How do Emotions Affect Esports Performance?

4.3.1 Positive and Negative Emotions. In this section, we explore RQ2a (*"What is the effect of positive emotions on performance?"*) and RQ2b (*"What is the effect of negative emotions on performance?"*). This section further dissects the literature on positive and negative emotions into approach tendencies, appraisal of emotions, and the effects of emotion on team performance.

Approach Tendencies. While previous research has uncovered what situations elicit positive and negative emotions, recent research efforts have explored how such emotions affect esports performance. In one such study, Behnke et al. [9] found that gamers displayed higher approach tendencies (i.e., characterised as a low or high tendency to approach or modify an emotional state) after watching enthusiastic and amusing videos (right before playing FIFA 19) compared to negative emotions and neutral conditions, thus improving performance (as measured by the number of goals scored)—approach behaviours (i.e., attempts to acquire desired goals) are said to promote better performance than avoidance behaviours (i.e., avoidance of distressing stimuli) in cognitive tasks [64]. Furthermore, enthusiasm was found to produce a stronger approach tendency than amusement, suggesting that the type of positive emotion also has an effect on performance outcomes. Interestingly, the elicitation of negative emotions (i.e., anger and sadness) had no effect on approach tendencies in comparison to the neutral conditions. However, work by Huang et al. [29] suggests that negative emotions improve performance in a time-constrained moving target selection task. Similar to Behnke et al. [9], emotions were elicited among participants by watching either positively (e.g., joy, tenderness, and amusement) or negatively (e.g., disgust and sadness) valenced videos before performing the task. Huang et al. [29] found that participants selected targets faster in the positive emotion condition; however, more mistakes were made. In contrast, targets were selected slower in the negative emotion condition, but fewer mistakes were made, resulting in better performance outcomes.

Appraisal. Recent work suggests that certain negative emotions, such as anxiety, may be helpful for maintaining optimal esports performance. A study by Schmidt et al. [68] investigated the influence of cortisol, flow, and anxiety on esports performance (assessed by win or loss). Salivary cortisol was measured before, during, and after competitive gameplay, while flow and anxiety were assessed via questionnaires immediately following a game. The findings revealed that winners experienced higher anxiety levels compared to losers, and that moderate levels of cortisol were associated with the highest performance and anxiety. These results indicate that a low to moderate physiological arousal alongside high levels of anxiety represent a favourable state for achieving optimal esports performance. In a similar vein, Wang et al. [79] investigated the relationship among gameplay self-efficacy, competition anxiety, and the performance of esports players who had participated in the Honor of Kings Champion Cup. In the context of this study, 'competition anxiety' refers to situation-specific forms of anxiety that result from fears about performance (categorised as negative appraisal anxiety, inability anxiety, and opponent performance anxiety). It was found that game self-efficacy was negatively related with competition anxiety, suggesting self-efficacy is a moderating factor of anxiety. In terms of performance, negative appraisal anxiety was positively related, whereas inability anxiety was negatively related. The positive relationship between negative appraisal anxiety and competition performance indicates that not all types of anxiety have a negative impact on performance. Wang et al. [79] suggest that external negative appraisals may stimulate a desire to win, thus leading to increased competition performance.

Team Performance. Research has also explored the role emotions play on esports performance at the team level. Abramov et al. [1] applied a machine learning approach to record the impact of emotions (via tones of communication) on team performance using audio recordings collected from a Team Fortress 2 tournament. While tones of communication may not explicitly represent an emotion, they may have an effect on other players' emotions. For instance, Abramov et al. [1] found a strong correlation between the performance of a team (as measured by kills, damage, assists, healing, and victories), communication between players and emotional sentiment (i.e., positive or negative) of communication. Teams that demonstrated positive tones (e.g., happiness) in communication were more successful in terms of performance compared to teams that used negative tones (e.g., anger) in communication. Additionally, team performance was further increased when teams engaged in more internal conversations. In sum, positive communication (and more of it) leads to improved overall team performance.

Summary. Combined, the current body of literature points to positive emotions (e.g., enthusiasm, amusement, joy, happiness) having a greater association with improved performance outcomes than negative emotions (e.g., anger and sadness) [1, 9]. However, there is not enough evidence that suggests negative emotions directly inhibit esports performance—some research has shown that negative emotions, such as certain types of anxiety actually play a role in improving esports performance [29, 68]; however, in other literature, such emotions have also been shown to have no significant impact on performance in either direction [9]. More research on the impact of negative emotions on esports performance is required to gain a better understanding of its effects. Similarly to Benke et al.'s [9] findings that enthusiasm had a stronger effect on the adoption of approach strategies than amusement, specific types of negative emotions may have increased levels of improved performance effects.

Taken together, it is inconclusive whether the emotion itself influences performance, or whether a third variable is responsible for the effects, such as the meaning that one assigns to a particular emotion. For instance, anger may lead to a defeated mindset in some individuals while others may use it as motivation to acquire desired goals. The outlook that a player adopts may be associated with the coping strategies they acquire through training. In esports, developing effective coping strategies is essential for regulating emotions and maintaining an optimal psychological state—an altered or negative emotional state may impact decision-making and overall gameplay, resulting in performance decrements [82]. As such, emotion regulation is essential for maintaining performance in competitive play and may moderate the effects of emotions on esports performance.

4.4 Category Overview: Emotion Regulation in Esports

Articles listed as 'emotion regulation' (n=15) are in line with RQ3, and focus primarily on how players regulate their emotions and the types of coping strategies they use during competitive play (see Table 2). These studies investigate emotional control among esports players and how they cope during play, revealing insights into the complex interplay between perceptions and actions (see sections 4.5.1 and 4.5.2). In terms of the methodologies used, six included quantitative measures, eight involved qualitative methods (e.g., interviews, focus groups, diary studies), and one was mixed-methods (i.e., interviews and online surveys)—see Figure 4.

Lastly, four articles comprised 'technical interventions' (see Table 3), two of which concern the development of technologies for the purpose of assisting esports players in regulating their emotions and improving performance. Both of these studies explored how biofeedback technologies can be used to improve player performance through experimentation; however, one also used a mixed-methods approach to evaluate usability (i.e., experiments and interviews). As seen in Figure 4, the remaining studies (n=2) constructed machine learning models designed to identify emotional states at specific points in time either through facial expressions or emotional speech patterns to assist players in regulating their emotions (see section 4.5.3).

4.5 RQ3: How do Players Regulate their Emotions?

4.5.1 Emotional Control. In this section, we explore RQ3a (*"Does emotional control play a role in esports performance?"*). In esports, emotional control is considered a core game competency among esports professionals [46]. The ability to overcome barriers of optimal performance in competitive play—such as ineffective attentional control, negative consequences of mistakes, going on tilt, dwelling on past performances, etc.—is essential for maintaining performance [27]. The following sections further breakdown the phenomena of tilt, mental toughness, and stress in relation to perceptions of emotional control.

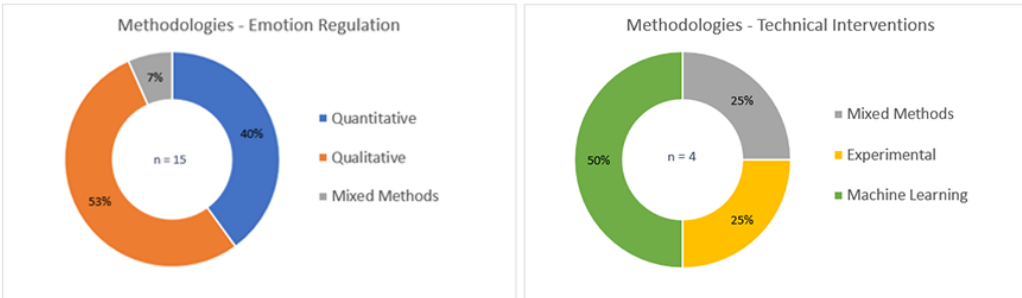

Fig. 4. Left: Methodologies used in studies related to emotion regulation in esports; Right: Methodologies used in studies related to technical interventions.

Tilt. Of particular relevance in the domain of esports in regard to emotion regulation is the phenomenon of 'going on tilt': a prolonged suboptimal state of mind occurring from a series of failures [81]. Once the point of tilt has been reached, it is difficult to reconcile performance as players often adopt the mentality that they have already been defeated. This sentiment is echoed in research by LeNorgant [40] that examined the role of self-talk in relation to performance in both traditional and online sporting contexts. It was found that esports competitors engage in less self-talk and have more negative thought patterns compared to traditional athletes, which contributes to the increased likelihood of experiencing tilt. Additionally, work by Wu et al. [82] examined player experiences of tilt in League of Legends and found that players who perceived tilt as malleable (i.e., within their control) were more likely to respond with positive strategies when the game was the cause of their tilt than those who perceived tilt as 'fixed' or unchangeable. However, perceptions of malleability made no difference in coping response when the players' own performance was responsible for their tilt. To reconcile performance, it is suggested that esports professionals would benefit from receiving early interventions targeted towards reframing players' approaches to tilt by implementing emotion regulation strategies that combat self-judgment, negative self-talk, and stress [40, 82].

Mental Toughness. Perceptions of emotional control have also been shown to be associated with mental toughness (i.e., a stress buffering personality trait) in traditional sports [33]—in particular, Kaiseler et al. [32] found that higher levels of mental toughness were associated with lower levels of perceived stress and higher levels of emotional control. However, these findings do not appear to translate to esports contexts. For instance, Poulus et al. [56] found that esports athletes with higher overall levels of mental toughness did not report lower levels of stress intensity or higher levels of stress control. Poulus et al. [56] suggest that this may be due to esports athletes experiencing different types of stressors compared to traditional athletes, such as technical issues and toxicity. Additionally, contrary to previous studies that have shown traditional athletes with higher mental toughness perceive stressors more as a challenge than a threat [47], findings by Poulus et al. [56] suggest that esports athletes appraise stressors as both a threat and a challenge. Poulus et al. [56] propose that the online nature of esports may contribute to uncertainty within non-professional play as players may not be able to choose their teammates.

Stress. Recent research by Madden and Harteveld [41] revealed that stress (particularly self-inflicted stress) had the greatest negative impact on the psychological wellbeing of esports players. Participating in highly competitive tournaments with limited rest between games often leads to stress among players. On top of that, esports players (particularly women) may also experience additional challenges such as toxicity or gender stereotyping [42]. Pursuant research by Madden et al. [42] explored how gender biases negatively impact female participation in esports. It was found that gender stereotypes persist within esports, namely that men are perceived as more ‘competitive’ while women are perceived as more ‘emotional.’ The stigma associated with being a ‘female gamer’ within a male-dominated scene likely contributes to increased stress among women and dissuades them from pursuing a career in esports [42]. As such, it is important to combat gender biases in esports by establishing an environment that is safe and inclusive for all.

Summary. Within the reviewed literature, there is consensus concerning the role emotional control plays in esports performance—that is, it is difficult to reconcile performance when the source of emotional distress is internal (e.g., individual performance failures) [41, 81, 82]. There is little speculation as to why internal sources of stress contribute to reduced feelings of emotional control; however, the basis for this finding may be rooted in a perceived external locus of control—appraising the root of performance failures as outside one’s control may serve as an avoidance coping mechanism to remove personal accountability, thus alleviating emotional distress associated with under performing. However, when confronted with a stressor within one’s control, players may feel helpless in alleviating emotional distress if they lack effective coping mechanisms. As such, it is important to equip players with effective coping strategies prior to competition as a preventative intervention. Furthermore, the literature suggests players would benefit from strategies that attempt to reframe one’s self perception, such as those targeted towards combating self-judgment, negative self-talk, stress, and gendered social stigma [40, 42, 82].

4.5.2 Coping Strategies. In this section, we explore RQ3b (*“What coping strategies do players use both on a team and on an individual level?”*). Developing effective coping strategies—such as social support, self-regulation, and psychological skill use—is crucial for maintaining an optimal state of mind during competitive play, and is associated with better performance outcomes [76]. In esports, players generally adopt a range of problem-focused (i.e., efforts to change a stressor), emotion-focused (i.e., efforts to regulate emotions), or avoidance-focused (i.e., efforts to disengage from the stressor) coping strategies. While players perceive problem-focused coping and emotion-focused coping as being more effective at reducing stress than avoidance coping strategies [58], research suggests that players tend to adopt more avoidance strategies during competitive play [39, 72]. Furthermore, problem-focused coping strategies have been found to be employed more often after competition [39]—it is suggested that equipping players with strategies designed to actively address the stressors they experience is a more adaptive approach for improving performance in the long-term than the use of avoidance strategies [56]. Studies have also found that the adoption of adaptive coping strategies is associated with mental toughness [56], less psychiatric and gaming disorder symptoms [6], as well as personality traits [70] The following sections further breakdown the coping literature in terms of how emotions are regulated in relation to the self, within teams, and their combined interplay at large.

Self Coping. Research by Kou and Gui found that LoL players regulate their emotions in relation to themselves (emotional self-care) and others (emotional leadership) during competitive play [36]. Emotional self-care involves internal strategies targeted towards improving the players’ own emotional wellbeing, while emotional leadership involves influencing the emotional states of others. Furthermore, Kou and Gui [36] identified five emotion regulation strategies commonly used among LoL players:

- Situation selection: approaching or avoiding certain situations (e.g., removing oneself from the game by taking a break).
- Situation modification: modifying the situation (e.g., muting players).

- Attentional deployment: shifting focus towards different goals or aspects of the game (e.g., letting go of the fear of losing).
- Cognitive change: reframing a situation to reflect a different perspective or meaning of the situation (e.g., "it's just a game").
- Response modulation: managing how one responds to a situation and expresses their emotions (e.g., refraining from typing negative comments in chat).
- Social interaction: turning to others in the community for emotional support (e.g., playing with friends).

Coping within Teams. Research has also explored emotion regulation in terms of building social support within esports teams. In one such study, Freeman and Wohn [22] interviewed 26 esports players to gain insight into the types of social support they experience from their gameplay. Freeman and Wohn [22] found that over time, esports players establish close interpersonal relationships with their teammates by which they receive support. Overall, four themes of social support were identified within esports teams: emotional support, informational support, instrumental support, and esteem support. Emotional support was mediated through friendships or romantic connections to team members. Informational support was provided through the transfer of game knowledge either by viewing how other team members play or by engaging in group discussions. Instrumental support refers to support received both within the game and outside of the game. For instance, team members supported the growth of each other as players within the game and also sought to support each other's personal goals outside of the gaming context. Lastly, esteem support was fostered within supportive team environments that were attentive to the emotional needs of others.

In another study, Smith et al. [72] identified the stressors and coping strategies of four elite CS:GO players during competitive play. The two primary types of stressors experienced by esports players are characterised as internal (team issues and individual issues) and external (scrutiny and criticism, and event issues). Team issues involve stressors such as communication issues, criticism from the in-game leader, lack of confidence in teammates, outcomes of losing, intra-team criticism, and lack of shared team goals. Individual issues are centralised around personal factors such as life balance and difficulties managing one's lifestyle. External issues refer to scrutiny and criticism received from the opposing team and social media as well as issues that arise from events such as the logistics and live audiences. Lastly, Smith et al. [72] identified that while players used a variety of coping styles such as emotion-focused (e.g., breathing) and problem-focused (e.g., intra-team communication) coping, players often reported an overuse of avoidance strategies during play (i.e., behaviours that avoid the stressor), and a lack of effective problem- and emotion-focused coping styles. This may be due to a lack of emotion regulation training among esports players, or it may be that the fast pace of esports does not facilitate the use of effective coping styles. Regardless, players would benefit from learning coping strategies that are effective in fast-paced and high-pressure environments. Likewise, this knowledge would be an asset to high school educational programs where players and esports staff identified the need for social-emotional learning interventions [16].

Interplay of Coping. Poulus et al. [57] qualitatively investigated the perceived determinants of success among 7 professional esports players—flow, focus, and confidence were reported when performing well, while breakdowns in team communication were identified as a major stressor. Pursuant research by Poulus et al. [58] found that performance-related stressors were more likely to be perceived as a challenge, while teammate-related stressors were more likely to be perceived as a threat. To facilitate team cohesion and prevent interpersonal breakdowns, players reported engaging in team-building activities within their teams [57]. In a similar vein, recent work by Hong and Connelly [28] explored the coping mechanisms used by high-performing esports players to enhance their physical and mental health throughout their careers. Through semi-structured interviews with 33 players, Hong and Connelly [28] identified three main themes: life balance, social support, and sleep management. Life balance was identified as crucial for enhancing health and wellbeing, and was maintained by taking breaks and engaging in other activities outside of esports. Social support from family, friends, or coaches was integral for promoting positive wellbeing, both during and after a player's career. Lastly, sleep management was recognized as a key coping mechanism essential for managing pressures associated with training loads and competition.

Summary. Taken together, the literature suggests that players adopt similar coping styles both at the individual and team level across various gaming contexts [22, 36, 72]. While players often identified problem- and emotion-focused coping as being more effective at reducing stress than avoidance-coping [56, 58], research shows that players tend to rely on more avoidance strategies during competitive play [39, 72]. This may be an artefact of the fast-paced nature of esports, which does not afford players adequate time to effectively apply problem-focused coping during competitive play. As such, players would benefit from receiving emotion regulation strategies effective in fast-paced, high-pressure environments—or, alternatively, a wider host of opportunities (built into competition formats) for the employment of more reflective coping styles. Furthermore, the literature suggests that individual issues outside of the gaming context (e.g., life balance, health and wellbeing, sleep management) translate into the team and game environment [28, 72]. As such, it is essential players receive effective coping strategies to manage their personal lifestyles to help ensure these factors do not carry over into the gaming context. Finally, fostering safe and inclusive environments within esports teams may also strengthen social support and improve self-esteem among teammates [22]. It is important to strengthen the emotional competencies of players in esports contexts as they are equally as important for maintaining optimal performance as the cognitive and sensory motor aspects of play [46].

4.5.3 Technical Interventions. In this section, we explore RQ3c (*"What technical interventions exist to assist players in regulating their emotions?"*). It is argued that esports contexts facilitate richer emotional experiences than traditional sports [36]. In addition to emotions evoked by socialising, teamwork, coordination, and communication (which are experienced in both contexts), esports players also experience emotions related to game design—an element that does not commonly affect traditional sports [36]. Although the technological nature of esports introduces more complexities in terms of emotions, it also offers potential for creating new technologies that support the development of emotion regulation strategies among esports players.

Acting on this potential avenue of research, researchers are already beginning to develop technological interventions in esports contexts. In one such study, Soler-Dominguez and Gonzalez [73] introduced a framework using virtual reality and neuroscience to evaluate psycho-cognitive characteristics of esports players with the goal of improving performance by creating customised training plans tailored towards specific player profiles. Each player profile is established based on six performance metrics: stress, engagement, interest, excitement, focus, and relaxation. Once a player profile is created, Soler-Dominguez and Gonzalez [73] suggest that applying neurofeedback training using real-time EEG data in a gamified environment can help adapt the brain activity of players to the optimum performance mode, thus improving performance. In another study, Drijfhout [17] developed a prototype of an informative stress feedback device to allow esports players to gain insight into their stress levels with the goal of allowing them to better regulate their emotions. The prototype includes four sensors used to generate stress biofeedback: electrocardiogram (ECG), galvanic skin response (GSR), Tilt, and Facial sensor. In an evaluation of the device during a game of Fifa, it was found that short bursts of stress are generated by events that occur close to a goal, and that stress levels rise near the end of the game compared to the beginning. Although the prototype requires further testing and possible design iterations, technical interventions have the potential to improve performance by providing more insightful feedback to players during gameplay. Lastly, researchers have also developed technical interventions for assessing emotions through analysing the facial expressions [31] and speech patterns of players [30]. In one such study, Hung and Chen [30] recently constructed an esports speech recognition model designed to assess emotional states of esports players and the corresponding performance in the game via speech recognition patterns. Using an efficient data training set, Hung et al. [31] developed an Esports Emotional Speech Recognition Model that can identify a player's mood response during specific points in time through speech recognition at 75 ± 2% accuracy. Hung et al. [31] suggest such a model could be used during post-game reviews to identify points in time that evoked emotional responses among players. This information could help players and coaches better identify areas for personal improvement.

Summary. Collectively, existing technical interventions are targeted towards providing individualised real-time feedback to players through various means (e.g., neuroscience, biofeedback, speech recognition) [17, 30, 73]. These personalised technical interventions recognise the unique needs of players—providing players individualised feedback with the goal of increasing awareness to issues that may have otherwise gone unnoticed. Technical interventions may be used in conjunction with one another or other training approaches to provide a more holistic understanding of each player's needs. As technical interventions continue to advance and become more accessible to esports players, the future of emotion regulation training looks promising for industry.

5 DISCUSSION

5.1 Summary of Results

Overall, the results suggest that while research in this area is developing (as evidenced by recently published work), more research is needed to better understand the role emotions play on esports performance and how players can effectively regulate their emotions. Specifically, we found that:

- Various in-game situations evoke emotional responses among esports players (e.g., achievement-related, teammate-related, game design-related, social identity-related).
- Emotions influence esports performance (positive emotions tend to improve performance compared to negative emotions); however, how one responds may be moderated by emotion regulation strategies.
- Esports players attempt to regulate their emotions during competitive play (typically employing avoidance coping styles), however, lack appropriate coping styles that effectively respond to the fast pace of esports contexts.
- Technical interventions are currently being developed to assist esports players in regulating their emotions and improving performance but require further development before their potential is realised in esports contexts.

In terms of the methodological approaches used within the literature, we found that:

- As seen in Figure 5, the majority of the studies used qualitative approaches (n=10), while the remaining included quantitative (n=8) approaches, experimental designs (n=6), mixed methods approaches (n=5), or machine learning models (n=3).
- Most of the studies included esports players as their sample (n=27), while the remaining used data available to the public through forums (n=1) or their sample was unspecified (n=4).
- Nine studies only included men in their sample, seventeen included a combination of both men and women, four were not applicable (e.g., studies that included scraper data where gender is unknowable), while the remaining two studies did not specify the genders of their sample (see Figure 5).

Table 1. Summary table of emotions and esports performance

Study	Research Focus	Methodology	Main goals of study
Mateo-Orcajada et al. [43]	Emotion and esports performance	5 male LoL players from the UCAM esports club completed questionnaires prior to competing and heart rate was measured during gameplay	To analyze differences in pre- and post-game mood as well as performance of esports players
Kou and Gui [36]	Emotion and esports performance, Emotion regulation	Conducted a thematic analysis on 431 posts from the LoL subreddit	To investigate how players experience and regulate emotions in LoL
Behnke et al. [8]	Emotion and esports performance	652 CS:GO players (617 male) wrote about situations in game that elicited positive and negative emotions	To identify which situations in gameplay elicit specific emotions
Behnke et al. [9]	Emotion and esports performance	241 male gamers watched video clips that either elicited positive emotions or negative emotions and then completed five matches of Fifa 19 while approach tendency, cardiovascular responses, and game scores were recorded	To determine which emotions improve performance and approach tendency
Huang et al. [29]	Emotion and esports performance	24 college students (12 males, 12 females) viewed either positive or negative video clips, then performed a time-constrained moving task	To explore the effects of induced emotion of the performance of a time-constrained movement task
Abramov et al. [1]	Emotion and esports performance	Audio recordings and game logs were collected from 7 Team Fortress teams during a tournament to train emotion recognition machine learning	To explore the impact of emotions on team performance using machine learning
Balint et al. [3]	Emotion and esports performance	450 participants (338 males, 112 females) were divided into four groups (non-gamers, LoL gamers, Fortnite gamers, LoL and Fortnite gamers), and completed the Anger Expression Scale	To assess aggression expression among LoL and Fortnite players
Haekal et al. [26]	Emotion and esports performance	EEG data from 10 participants (5 male gamers, 5 male non-gamers) was recorded while playing PUBG Mobile to assess levels of stress experienced throughout different gameplay phases	To evaluate levels of stress when playing PUBG Mobile
Mavromoustakos-Blom et al. [44]	Emotion and esports performance	Examined 17 Hearthstone players during competition to determine whether eye gaze, head pose, and facial expressions are associated with subjective reports of emotion	To investigate whether objective data from face recordings are associated with subjective player experiences
Schmidt et al. [68]	Emotion and esports performance	Salivary cortisol of 23 players (19 males, 4 females) was recorded before and after competitive play, and players completed the Flow Short Scale after their games	To examine the relationship between physiological arousal, anxiety, flow, and performance during competition

Study	Research Focus	Methodology	Main goals of study
Shin et al. [71]	Emotion and esports performance	12 male participants underwent EEG recordings continuously during and post Counterstrike gameplay, and provided subjective ratings of their emotional responses pre-, during, and postgame	To explore the emotions associated with Counterstrike gameplay
Wang et al. [79]	Emotion and esports performance	232 players (150 males, 82 females) in the Honor of Kings Champion Cup completed questionnaires related to gameplay self-efficacy and competition anxiety	To assess the relaitonship among gameplay self-efficacy, competition anxiety, and the performance of esports players
Piton et al. [53]	Emotion and esports performance	26 participants (19 males, 7 females) played two games of Hearthstone while their affective state was monitored in real time through self-reporting and facial expression recognition	To determine whether peak-end effects occur in gaming experiences

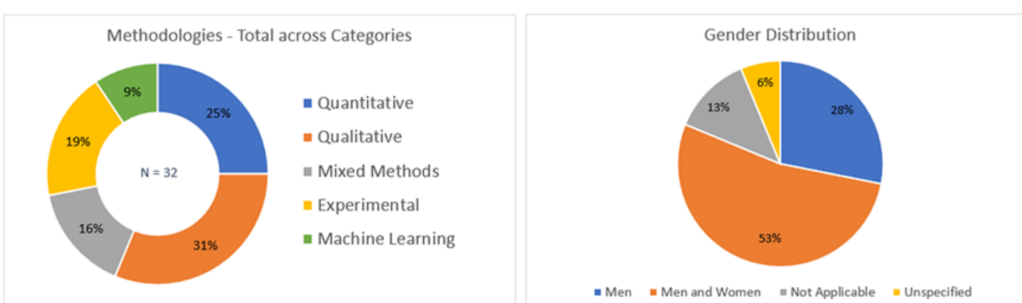

Fig. 5. Left: Methodologies used across all study categories included in the review; Right: Gender distribution of participants across all studies included in the review.

5.2 Implications and Future Work

This systematic review represents an initial step in better understanding how emotions influence esports performance and how players regulate their emotions. The findings from the review reveal implications for researchers, game designers, and esports personnel (e.g., coaches, managers, players).

For researchers, providing a comprehensive review is helpful for understanding the current state of the literature and identifying potential avenues for future research. For instance, there appears to be a lack of research that experimentally investigates the effect of emotions on esports performance. Conducting future research using experimental approaches will help infer causality concerning how emotions affect esports performance. Additionally, using more mixed-methods approaches may also provide a more complete understanding of the role emotions play on performance. Future work should also endeavour to include marginalised groups in their samples, and explore potential differences in emotional responses and effects on performance. For example, the experience of play often differs between men and women (as women are disproportionately the victims of in-game harassment [20]); to avoid potential fallacies or assumptions that emotions are experienced the same way, researchers should consider demographic contexts. Furthermore, the effect of

Table 2. Summary table of emotion regulation on esports performance

Study	Research Focus	Methodology	Main goals of study
Wu et al. [82]	Emotion regulation	95 esports players (90.7% men) from a high school esports league provided descriptions of their experiences and responses to tilt as well as their perception of its malleability	To determine esports players' perceptions of tilt and its malleability
Poulus et al. [56]	Emotion regulation	316 (90% men) esports players (ranked in the top 40% of a major esport) completed questionnaires	To explore stress and coping in esports athletes and the influence of mental toughness
Madden and Harteveld [41]	Emotion regulation	Used a mixed-methods approach by conducting semi-structured interviews with 10 esports professionals (8 males), and an online survey with 68 esports players (53 males)	To identify the biggest health concerns among esports players
Madden et al. [42]	Emotion regulation	19 professional gamers and event organizers (10 females, 9 males) were interviewed about their personal experiences with gender biases in esports	To investigate how gender biases negatively impact female participation in esports
Freeman and Wohn [22]	Emotion regulation	26 esports players (85% men) were interviewed about their experiences of social support in competitive digital environments	To explore what types of social support esports players experience from their gameplay
Smith et al. [72]	Emotion regulation	7 male CS:GO players who had recently participated in a competitive event were interviewed about stressors they experience during play as well as the coping strategies they use	To examine the stressors faced by esports players during competitive play as well as the coping strategies they use
Nagorsky and Wiemeyer [46]	Emotion regulation	1,835 esports players (95% male) completed online questionnaires which were used to reveal relevant game competencies and training areas	To identify the gaming competencies of esports players and investigate training regimes relevant to different esports
LeNorgant [40]	Emotion regulation	33 esports players (91% men) and 56 traditional sport athletes (88% men) completed two sets of questionnaires: once after initial recruitment and again within 24 hours of competition	To compare the intensity and frequency of self-talk patterns among esports players and traditional sport athletes
Banyai et al. [6]	Emotion regulation	3476 gamers (3133 males) completed questionnaires concerning psychiatric symptoms, coping strategies, and symptoms of gaming disorder	To explore the moderating effect of coping strategies on the relationship between psychiatric symptoms and gaming disorder among gamers
Cho et al. [16]	Emotion regulation	Conducted focus groups with 38 student esports teams from 25 highschools with LoL players (39; 90% men), teachers (11), and coaches (5), as well as interviews with parents (10)	To determine the building blocks of a successful highschool esports program

Study	Research Focus	Methodology	Main goals of study
Hong and Connelly [28]	Emotion regulation	33 (97% men) esports players (21 professional, 6 semi-professional, 4 amateur, and 2 retired) were interviewed about their coping skills and strategies	To investigate the coping strategies used by esports players to enhance their physical and mental health
Poulus et al. [57]	Emotion regulation	7 male professional esports players were interviewed about their perceptions of success determinants in esports performance	To explore the perceived determinants of success in professional esports players
Poulus et al. [58]	Emotion regulation, Emotion and esports performance	6 professional male LoL players completed weekly diaries over a competitive season (87 days) documenting experienced stressors, stressor intensity, coping strategies used, and perceived coping effectiveness	To longitudinally examine the stressors, stress appraisal, coping, and coping effectiveness experienced by professional players
Semenova et al. [70]	Emotion regulation	34 male esports players (16 professional, 18 amateur) completed the IPIP-NEO Big Five personality traits questionnaire as well as the Ways of Coping Questionnaire (WCQ)	To assess the relationship between personality traits and coping strategies of esports players
Trotter et al. [76]	Emotion regulation	1,444 esports players completed questionnaires to social support (Athletes Received Support Questionnaire), self-regulation (Self-Regulation Questionnaire), and psychological skill use (Test of Performance Strategies-2)	To determine if self-reported social support, self-regulation, and psychological skill use influences in-game rank.

Table 3. Summary table of technical interventions for esports performance

Study	Research Focus	Methodology	Main goals of study
Soler-Dominguez and Gonzalez [73]	Technical interventions	Used virtual reality and neuroscience to develop player profiles by evaluating psycho-cognitive characteristics of players	To improve performance in esports by using player profiles and neurofeedback within a gamified training environment
Drijfhout [17]	Technical interventions	3 participants (unspecified) tested the prototype and provided usability feedback via interviews	To develop a prototype that provides stress feedback to esports players to help regulate emotions and improve performance
Hung et al. [31]	Technical interventions	2400 images of facial expressions (happy, angry, sad, etc.) were used to train the model through transfer learning	To construct a gaming facial emotion recognition model to identify the emotions of players and assist in esports training interventions
Hung and Chen [30]	Technical interventions	Implemented a speech emotion recognition training model involving audio input, audio processing, audio feature extraction, and data training	To develop an esports speech recognition model designed to identify emotional points in the game where players could work on regulating during training

negative emotions on esports performance is unclear, with some research suggesting that certain types of negative emotions (e.g., anxiety) actually play a role in improving performance [29, 68]. Future research exploring the role of negative emotions on performance may help inform this understanding—in line with work by Behnke et al. [9], exploring specific types of negative emotions (e.g., sadness, anger, frustration, anxiety) may further our understanding of which types of negative emotions yield the most optimal performance outcomes.

In terms of further exploration in the field of affective computing, research should continue to explore the development of technological interventions designed to assist esports players in managing their emotions and improving performance. Current approaches utilise real-time feedback through various means (e.g., neuroscience, biofeedback, speech recognition) [17, 30, 73] to provide players an individualised assessment of their emotions in relation to performance. However, research has yet to evaluate the efficacy of such approaches when used in conjunction with one another. While it is important to be mindful of the limitations of using various physiological measures simultaneously (e.g., participant comfort, performance interference), it may provide a more holistic understanding of each player's needs if executed appropriately. Incorporating technological interventions into esports training and management would revolutionise the way we prepare players to perform their best.

For game designers, understanding how aspects of game design influence emotional responses among players is important for making design decisions. For instance, lack of game clarity or changes in game balance (e.g., character buffs or nerfs) may elicit frustration among players and, in turn, decrease performance and enjoyment[36]. Adding to the complexity of game design are the challenges of anonymity and online disinhibition in online gaming, which have been linked to toxicity [11, 34, 75]. Toxicity is a pervasive issue in online gaming disproportionately affecting women [20], LGBTQ players [4], and players of colour [23], which has been shown to affect women's participation in esports [42]. As such, game designers should take toxicity into account when implementing game features such as voice chat among random teammates and weigh the potential pros (e.g., team communication) and cons (e.g., targeted harassment) accordingly. Furthermore, understanding how players manage their emotions during competitive play may inform game designers of ways to allow players to cope. For instance, while players report problem- and emotion-focused coping as being more effective [56, 58], they tend to use more avoidance styles of coping in game [39, 72]. Game designers could assist players in adopting more problem-focused styles by reminding them to communicate and strategize with teammates, uplift one another when team performance is suffering, or make personal performance goals (e.g., trying not to die repeatedly in a specific time frame). Players also report performance being more difficult to reconcile when their poor performance is the source of their frustration [36, 41, 82]. As such, implementing problem- or emotion-focused coping interventions targeted towards alleviating personal stress during breaks in gameplay (e.g., a positive message during a death screen, a reminder to take a break in between matches) may inform players how to cope with their emotions. These breaks in play represent key opportunities for designers to implement a range of grounding techniques for emotional deescalation (e.g., asking players to identify objects in their surroundings, playing soothing music during death screens). However, game designers must practice caution in implementing such features in-game to avoid disrupting the player experience.

For esports training and management, this knowledge may be helpful for informing training interventions designed to regulate emotions. More specifically, training regimes that include technological interventions that inform players of their emotional states may help players better regulate their emotions during play. While current technologies can be intrusive to play (e.g., EEG attachments), future interventions could implement such technologies more seamlessly into existing equipment for a less interrupted experience. For instance, a gaming mouse that measures pulse and vibrates when a player has an elevated heart rate could help inform the player it is time to relax and take a deep breath; a gaming headset with EEG attachments that relays a player's emotional states to management could inform practice sessions; or a gaming chair that records how long a player has been stationary and sets reminders for daily physical activity could improve overall health and well-being. Additionally, esports players may benefit from offline instruction that teaches effective coping strategies to use during play. Currently, players are unable to effectively cope during gameplay likely due to the fast pace of the game [72]—as such, brief emotion-focused coping strategies such as taking a deep breath may provide the necessary temporary relief. In offline settings, players may benefit from hiring an esports psychologist to guide them through more problem-focused coping strategies to develop effective coping styles in the long-term, as well as lifestyle management (e.g., sleep management, work-life balance) for optimal wellbeing.

Further, esports promises translational learnings and opportunities in other domains. While esports proffers high pressure scenarios that demand excellence in performance, they do so in a physically safe environment—and, in non-professional contexts, in environments that do not represent significant stakes for their players. As such, we contend that there may be far-reaching benefits in employing esports as a vehicle for training emotion regulation—for example, as an intervention tool for people who struggle with appropriate coping strategies. This runs parallel to the widespread and successful adoption of 'serious games' in vocational training and rehabilitation [61, 80].

5.3 Limitations

While we accomplished our goal of identifying the current state of the literature concerning the role emotions play in esports performance and how players regulate their emotions, this review is not without its limitations. We acknowledge that the key words used within our search strategy may not have encompassed all the literature relevant to the area of interest; for example, esports titles with more diminutive tournament prize pools (e.g., Vainglory, Street Fighter)—or who may have been more dominant over five years ago (e.g., Quake, Doom)—were not directly included in the key words. As such, relevant articles may have been missed in the initial search; however, snowballing additional sources cited within key articles may have overcome this limitation. Furthermore, we excluded literature examining child populations from our review; this was motivated by the differences present in the experience of emotions between child and adult populations, in which children and adolescents' emotion regulation is heavily influenced by developmental trajectory—such as the current state of cognitive or socioemotional development [66]. We contend that a review specifically investigating children's emotions in esports contexts would be valuable, especially as youth esports leagues see increased uptake within K-12 schools.

6 CONCLUSION

In this systematic review, we identified the state of the literature concerning the role emotions play in esports performance as well as highlighted coping strategies players use to regulate their emotions during competitive play. Additionally, we briefly reviewed the potential of technical interventions in esports training for regulating emotions among players. Overall, we found that gameplay situations elicit a variety of emotions that impact esports performance. Furthermore, we found that esports players attempt to regulate their emotions during competitive play; however, lack appropriate coping styles that effectively respond to the fast pace of esports contexts. Moreover, technical interventions are currently being developed to assist esports players in regulating their emotions. We provide implications for researchers, game designers, and esports personnel and suggest potential avenues for future research. Research that continues to examine the role emotions play on esports performance will help support the emotional wellbeing of players as well as promote the longevity of esports.

7 OTHER INFORMATION

Provided per PRISMA 2020 guidelines [49].

7.1 Registration and Protocol

This protocol for this review was not registered.

7.2 Support

No financial or non-financial support was provided for this review.

7.3 Competing Interests

The authors of this study report no competing interests or sponsors that could have influenced the protocol or reporting of this review.

7.4 Availability of Data, Code, and Other Materials

The full corpus of included literature, including the key variables extracted from this literature, are included in Tables 1, 2, and 3.

ACKNOWLEDGMENTS

Thank you to Julian Frommel for providing expert guidance and feedback on this work.

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Received 2023-02-21; accepted 2023-07-07